



[GAME NAME HERE]: An Educational Bubble-shooter game centered around Changes in Matter

A Capstone Project

Presented to

The Department of Software Technology
College of Computer Studies
De La Salle University

In partial fulfillment

Of the requirements for the degree of

Bachelor of Science in Interactive Entertainment
Major in Game Development

and

Bachelor of Science in Interactive Entertainment
Major in Game Art and Design

BALDONADO, Matthew Jayd F.

MONCADA, Julianna Mikaela R.

SALANGA, Francesca Paula P.

SY-WICO, Juan Morcwel D.

UBAY, Conrad Amadeus Carmelo C.

DIMAUNAHAN, Ryan
Adviser

December 5, 2025

ABSTRACT

Chemistry education contains foundational concepts that are often perceived as difficult for students to understand, particularly in terms of their application to real-world situations. This difficulty may lead to cognitive overload, especially when instruction lacks opportunities for practical engagement. Interactive learning approaches allow students to actively apply and develop their analytical skills while discovering how chemistry concepts relate to everyday experiences.

This study explores the use of Game-Based Learning (GBL) through the development of [GAME NAME HERE], an educational serious game designed as a learning material for Grade 6 students under the MATATAG curriculum in the Philippines. The game focuses on physical and chemical changes of matter, expanding students' understanding beyond the three common states of matter. Through interactive gameplay, students engage with concepts such as phase changes and heat transfer while learning to distinguish physical changes from chemical changes.

The primary goal of this study is to improve students' conceptual understanding, motivation, and emotional engagement by reinforcing hands-on learning and real-world applications of chemistry concepts. This project aims to provide both students and teachers with an accessible and engaging instructional resource that supports visualization, experimentation, and the development of analytical skills in basic chemistry education.

Keywords: physical and chemical changes, game-based learning, educational serious game

Sustainable Development Goals: 4: Quality Education, 11: Sustainable Cities and Communities, 12: Responsible Consumption and Production

ABSTRAK

Ang pag-aaral ng kimika ay naglalaman ng mga pundamental na konsepto na kadalasang itinuturing na mahirap unawain ng mga mag-aaral, lalo na pagdating sa aplikasyon nito sa totoong buhay. Ang ganitong kahirapan ay maaaring magdulot ng cognitive overload, partikular kung kulang ang mga aralin sa praktikal at interaktibong gawain. Sa pamamagitan ng interaktibong pagkatuto, aktibong nahahasa ng mga mag-aaral ang kanilang kasanayan sa pagsusuri at mas nauunawaan nila kung paano nagagamit ang mga konsepto ng kimika sa pang-araw-araw na buhay.

Sinisiyasat ng pag-aaral na ito ang paggamit ng Game-Based Learning (GBL) sa pamamagitan ng pagbuo ng [GAME NAME HERE], isang pang-edukasyong seryosong laro na idinisenyo bilang pantulong na materyal sa pagkatuto para sa mga mag-aaral sa Baitang 6 sa ilalim ng MATATAG kurikulum sa Pilipinas. Nakatuon ang laro sa pisikal at kemikal na pagbabago ng materyal, na nagpapalawak sa kaalaman ng mga mag-aaral lampas sa tatlong karaniwang kaanyuan ng matter. Sa pamamagitan ng interaktibong gameplay, natututuhan ng mga mag-aaral ang mga konsepto tulad ng pagbabago ng estado at paglipat ng init, habang malinaw na natutukoy ang kaibahan ng pisikal at kemikal na pagbabago.

Layunin ng pag-aaral na ito na mapabuti ang pag-unawa, motibasyon, at emosyonal na pakikipag-ugnayan ng mga mag-aaral sa pamamagitan ng hands-on na pagkatuto at aplikasyon ng mga konsepto ng kimika sa mga sitwasyong hango sa totoong buhay. Nilalayon din ng proyektong ito na magsilbing kapaki-pakinabang na materyal para sa mga guro at mag-aaral upang mas mapadali ang pagkatuto ng kimika sa isang biswal, interaktibo, at makabuluhang paraan.

Mga susing salita: pisikal at kemikal na pagbabago, game-based learning, pang-edukasyon na laro

Contents

Abstract	i
Abstrak	ii
1 INTRODUCTION	1
1.1 Background of the Study	1
1.2 Statement of the Problem	2
1.3 Project Objectives	2
1.4 Scope and Limitations	2
1.5 Significance of the Project	3
2 REVIEW OF RELATED LITERATURE	4
2.1 Literature Review on the Educational Challenges in the Philippines in both a General and Scientific Scope	4
2.2 Literature Review on Educational Serious Games	5
2.3 Literature Review on Chemistry Game-Based Learning and Existing Chemistry Games	5
2.4 Literature Review on Bubble Shooter Games for Learning	6
3 GAME DESIGN	7
3.1 Game Name	7
3.2 Game Overview	7
3.3 Game-Concept	7
3.4 Genre	7
3.5 Target Audience	8
3.6 Game Flow Summary	8
3.7 Look and Feel	8
3.8 How does the game insert itself in a pedagogical scenario?	8
3.9 Gameplay and Mechanics	9
3.9.1 Gameplay	9
3.9.2 Moment to Moment	9
3.9.3 Mechanics	9

3.9.4	Physics	10
3.9.5	Movement	10
3.9.6	Objects	10
3.9.7	Actions	10
3.9.8	Combat	10
3.9.9	Economy	10
3.9.10	Game Options	10
3.9.11	Constraints due to the pedagogical objectives	11
3.10	Story, Setting and Character	11
3.10.1	Story and Narrative	11
3.10.2	Game progression	11
3.10.3	License considerations	11
3.10.4	Game World	11
3.10.5	Characters	11
3.10.6	Levels	11
3.10.7	Training levels	12
3.10.8	Assessment	12
3.11	Interface	12
3.11.1	Visual System	12
3.11.2	Control System	12
3.11.3	Audio, music, sound effects	12
3.11.4	Help System	12
4	DESIGN AND IMPLEMENTATION	13
4.1	Design Changes	13
4.2	Implementation Challenges	13
4.3	Formative User Testing Observations	13
5	METHODOLOGY	14
5.1	Research Design	14
5.2	Data Collection	14
5.3	Data Analysis	14
5.4	Ethical Considerations	14
	REFERENCES	15

List of Figures

List of Tables

2.1	Summary Table of Papers about the Educational Challenges in the Philippines .	4
2.2	Summary Table of Papers about Science Education in the Philippines	4
2.3	Summary Table of Papers about Educational Serious Games	5
2.4	Summary Table of Papers on Chemistry game-based learning	6
2.5	Summary Table of Papers on Stoichiometry Learning	6
2.6	Summary Table of Papers on Bubble Shooter Games	6
3.1	Game Loop Table	9

CHAPTER 1

INTRODUCTION

1.1 Background of the Study

A strong foundation in chemistry is essential for students early in their academic careers, as it enables them to understand more complex concepts and recognize their applications in real-world contexts. However, many students struggle with chemistry due to the abstract nature of its foundational topics and the limited opportunities to apply these concepts in practical settings.

One major factor contributing to student disengagement and cognitive overload is the structure of traditional learning materials, which often emphasize immediate mastery rather than gradual understanding. In many classroom environments, students are expected to apply large amounts of newly introduced information without sufficient time to practice or explore concepts meaningfully (Saputri et al., 2025). As a result, chemistry is frequently perceived as difficult and unapproachable, leading to reduced motivation and interest among learners.

A hands-on and interactive approach to learning can help students better internalize abstract concepts by transforming them into tangible experiences (Bridges, 2015). Interactive learning environments encourage active participation, allowing students to develop critical thinking and analytical skills through experimentation and problem-solving. Additionally, immediate feedback supports the correction of misconceptions, enabling students to reflect on and improve their understanding in real time (Blyznyuk & Kachak, 2024).

These factors significantly influence students' motivation and ability to engage with foundational chemistry topics, particularly those involving changes in matter. Before students can fully grasp phase changes, they must first master prerequisite concepts. However, rigid curriculum pacing often prevents learners from fully absorbing new ideas before progressing to subsequent topics, resulting in frustration and disengagement (Treagust et al., 2018).

Given these challenges, there is a clear need for learning approaches that promote active engagement without overwhelming students. Instructional tools that balance accessibility, relevance, and interactivity can enhance comprehension while supporting the development of cognitive and critical thinking skills (Ren et al., 2024). This study aims to develop a curriculum-aligned approach that serves as a supplementary learning tool, helping students transition from disinterest to engagement while fostering a more positive emotional experience with chemistry learning.

Furthermore, integrating a Game-Based Learning (GBL) approach as a supplement to traditional instruction may strengthen students' conceptual understanding and analytical abilities (Putri et al., 2022). By embedding chemistry concepts within interactive gameplay, the study seeks to support long-term learning and improve students' confidence in engaging with foundational chemistry topics.

1.2 Statement of the Problem

A key challenge in Grade 6 chemistry education is bridging the gap between learning scientific concepts and applying them to real-world situations. While students are taught about changes in the state of matter (Department of Education, 2024), these concepts are often perceived as abstract and disconnected from everyday experiences. As a result, learners may struggle to recognize the relevance and practical value of chemistry knowledge outside the classroom. This disconnect has been associated with poor knowledge retention and difficulty applying scientific concepts in authentic contexts among Filipino students (Barrot et al., 2021).

Traditional instructional methods tend to prioritize textbook-based learning and factual recall, often overlooking the importance of contextual understanding and experiential learning. Consequently, students may understand what happens during a physical or chemical change but fail to grasp why it occurs or how it applies to real-life scenarios.

[GAME NAME HERE] addresses this challenge by serving as a curriculum-aligned learning tool that integrates chemistry concepts into interactive, real-world-inspired scenarios. This study investigates whether embedding MATATAG curriculum competencies into gameplay that simulates practical situations can improve student retention, engagement, and conceptual understanding. By transforming academic concepts into tools for problem-solving within the game, students are encouraged to learn through application, experimentation, and reflection.

1.3 Project Objectives

The main objective of this project is to develop and evaluate a Game-Based Learning (GBL) tool that supports the effective teaching of chemistry concepts under the Grade 6 MATATAG Science curriculum. Specifically, the study aims to:

1. Design a game that supports the learning outcomes required under the Grade 6 First-Quarter Science syllabus of the MATATAG curriculum, including topics such as changes in the state of matter, physical and chemical changes, and mixtures and separation techniques (Department of Education, 2024).
2. Develop [GAME NAME HERE] as an interactive learning tool that reinforces curriculum content while demonstrating the situational and real-world application of chemistry concepts.
3. Evaluate whether a Game-Based Learning (GBL) approach using [GAME NAME HERE] can improve student performance, interest, and knowledge retention.

1.4 Scope and Limitations

This study focuses on the development of [GAME NAME HERE] as a curriculum-aligned educational game designed to support the Grade 6 MATATAG First-Quarter Science syllabus

(Department of Education, 2024). The game specifically targets learning outcomes related to changes in the state of matter, physical and chemical changes, and mixtures and separation techniques. Its bubble-shooter mechanics are intended to help students visualize chemical processes through interactive elements while encouraging strategic thinking. To prevent cognitive overload, the game limits its content to topics included in the official curriculum.

The game bridges classroom instruction and real-world application by embedding chemistry concepts into familiar scenarios, such as food preparation and household activities. Actions such as slicing, crushing, heating, and cooling are used to illustrate physical and chemical changes in a simplified manner appropriate to a Grade 6 learner's cognitive level. To maintain accessibility, the game avoids introducing excessive variables that may overwhelm students.

The study evaluates the game's effectiveness through pretests and posttests targeting specific competencies, as well as engagement metrics collected during gameplay. Assessment tasks are aligned with MATATAG performance standards, such as designing fair tests to compare physical changes. However, the study is limited by its duration and does not assess the long-term transfer of learning to future chemistry topics beyond the scope of the game.

1.5 Significance of the Project

This project explores the potential of educational serious games as a learning tool for teaching abstract and foundational science concepts, particularly in Grade 6 chemistry. With the implementation of the MATATAG curriculum, there is a growing need for innovative instructional materials that align with its revised structure and learning priorities. This curriculum shift changes how students are introduced to topics such as states of matter, physical changes, and basic chemical interactions.

By providing an interactive and gamified learning environment, [GAME NAME HERE] allows students to visualize and manipulate abstract chemistry concepts through experiential learning. The game's design breaks down complex ideas into manageable segments, supporting better comprehension and retention through repetition and hands-on engagement.

Additionally, the project offers practical support for teachers and learners adapting to the MATATAG curriculum. It supports differentiated instruction by addressing visual and kinesthetic learning styles while encouraging independent exploration and critical thinking. Beyond its immediate educational impact, the study contributes to the broader field of educational game development, particularly within the Philippine context where access to high-quality, engaging science learning materials remains limited. As such, [GAME NAME HERE] advances research on game-based learning while responding to the evolving needs of science education in a developing educational system.

CHAPTER 2

REVIEW OF RELATED LITERATURE

2.1 Literature Review on the Educational Challenges in the Philippines in both a General and Scientific Scope

The table below provides a summary of the research papers that discussed the status of the educational system in the Philippines. It covers the issues with the K to 12 program, implementation of the MATATAG curriculum, and the effectiveness of the hybrid learning setup.

Paper Title	Methodology	Findings	Recommendations
-------------	-------------	----------	-----------------

Table 2.1: Summary Table of Papers about the Educational Challenges in the Philippines

The K–12 program, which is the current educational framework in the Philippines, has faced multiple challenges, particularly curriculum congestion (Kilag et al., 2024). The large volume of topics and activities required within limited instructional time places significant pressure on both teachers and students. Educators often struggle to cover all mandated competencies effectively, while students experience increased academic load and stress. Additional challenges include limited instructional resources and shortages of adequately trained teachers.

In response to these concerns, the Department of Education introduced the MATATAG Curriculum, which aims to address curriculum congestion and better respond to learners' needs. The MATATAG Curriculum emphasizes critical thinking, digital literacy, and global competence. However, as the curriculum is still in its early stages of implementation, continuous evaluation is necessary to assess its effectiveness and identify areas for improvement.

Despite the easing of pandemic restrictions, many schools continue to employ hybrid learning approaches. Research indicates that students' experiences with online and blended learning vary depending on factors such as access to resources, quality of teacher interaction, and socioeconomic conditions (Barrot et al., 2021). Studies have shown that challenges in online learning environments are more commonly associated with home learning conditions rather than technological literacy, highlighting the need for accessible and flexible instructional tools.

The table below is a summary of the research papers that tackle the situation of Science Education in the Philippines. It covers the situation of chemistry learning in the Philippines and the new science curriculum that is being planned for implementation.

Paper Title	Methodology	Findings
-------------	-------------	----------

Table 2.2: Summary Table of Papers about Science Education in the Philippines

In science education, curriculum reform has been driven by issues such as mismatched prerequisite knowledge and consistently low performance in national and international assessments. Under the MATATAG framework, science education aims to strengthen students' technological and scientific literacy to meet the demands of the 21st century and emerging industries (Diquito, 2024). Studies on chemistry instruction indicate that student performance and engagement improve when instructional approaches incorporate interactive strategies such as computer simulations, inquiry-based learning, cooperative learning, and project-based e-learning (Vallespin, 2024; Vega, 2024).

2.2 Literature Review on Educational Serious Games

The table below showcases the papers that cover educational serious games. It discusses the impact of serious games and use of outcome-based methodology for game design.

Paper Title	Methodology	Findings
-------------	-------------	----------

Table 2.3: Summary Table of Papers about Educational Serious Games

Existing literature distinguishes serious games from gamification. Gamification refers to the integration of game design elements into non-game contexts, while serious games are complete digital games designed with both educational and entertainment goals. Research suggests that both approaches can positively influence learning outcomes, though they affect motivation differently. Gamification tends to enhance extrinsic motivation, whereas serious games are more strongly associated with intrinsic motivation (Ren et al., 2024).

For serious games, outcome-based development methodologies are commonly recommended. In such approaches, learning objectives serve as the foundation for designing gameplay mechanics, narratives, and feedback systems (Sison et al., 2018). This alignment ensures that educational goals remain central to the player experience and that gameplay directly supports learning outcomes.

2.3 Literature Review on Chemistry Game-Based Learning and Existing Chemistry Games

The tables under this category contain a summary of various papers that discuss the other existing chemistry games. The papers under this category also discuss the effects of game-based learning and struggles in studying stoichiometry.

Paper Title	Methodology	Findings
-------------	-------------	----------

Table 2.4: Summary Table of Papers on Chemistry game-based learning

Game-Based Learning (GBL) promotes both hands-on and minds-on learning, leading to deeper conceptual understanding (Byusa et al., 2022). Research has shown that digital educational games can improve critical thinking skills, learning retention, and student motivation by presenting content in an engaging and interactive manner (Putri et al., 2022).

Many existing chemistry educational games are puzzle-based, such as Sokobond Express and Element Solitaire (Elementeo), while others adopt role-playing or simulation-based formats (Hu et al., 2022).

Paper Title	Methodology	Findings
-------------	-------------	----------

Table 2.5: Summary Table of Papers on Stoichiometry Learning

Studies also highlight that difficult chemistry topics, such as stoichiometry, often benefit from game-based approaches due to their ability to increase motivation and reduce cognitive barriers. Even non-digital educational games, such as board games modeled after snakes and ladders, have demonstrated positive effects on students' motivation and learning outcomes in chemistry-related topics (Janah & Mitarlis, 2024).

2.4 Literature Review on Bubble Shooter Games for Learning

The table below showcases the research papers on bubble-shooter type games. The papers under this category explore how effective a bubble-shooter mechanic is for learning.

Paper Title	Methodology	Findings
-------------	-------------	----------

Table 2.6: Summary Table of Papers on Bubble Shooter Games

Bubble shooter games have been found to support memorization and visualization due to their repetitive and pattern-based mechanics (Maung, 2015). Research-oriented bubble shooter games such as Bias Blaster and Chemistry Pop, though not commercially published, demonstrate the potential of this genre as an educational tool. These games show that bubble shooter mechanics can effectively support learning by reinforcing repetition, visual association, and strategic thinking.

CHAPTER 3

GAME DESIGN

3.1 Game Name

The title of the game is [GAME NAME HERE]. The name reflects the game's overarching narrative, which centers on a neighborhood chemist helping community members with everyday challenges. It also incorporates wordplay, as the term "bubbly" refers both to the character's personality and to the game's bubble-shooter mechanics.

3.2 Game Overview

[GAME NAME HERE] is a 2D bubble-shooter educational game designed to teach Grade 6 chemistry concepts through interactive gameplay. Players create and transform materials using five action orbs, each representing a specific physical or chemical process. In addition to bubble-shooter levels, players can explore a village environment and assist characters through a sorting minigame focused on identifying salvageable and unsalvageable materials.

3.3 Game-Concept

[GAME NAME HERE] is a 2D stylized educational game set in a brightly colored neighborhood and follows a chemist as she assists the community with daily tasks. The challenges focus on core chemistry topics such as states of matter and, more importantly, distinguishing between physical and chemical changes in everyday life. In early levels, particularly those set in a café, players use action orbs on ingredients to observe changes in properties and achieve goals defined by a recipe system.

Progress is communicated through a traffic-light feedback system, where red indicates no progress, yellow indicates partial progress, and green indicates completion. Players must strategically use action orbs such as cooling, heating, mixing, slicing, crushing, and swapping to achieve correct outcomes for each level.

3.4 Genre

The game combines educational gaming with puzzle and simulation elements. The bubble-shooter mechanics place it within the puzzle genre, emphasizing logic, accuracy, and planning. The neighborhood exploration component functions as a light simulation, reinforcing learning through contextual storytelling and interaction. This hybrid genre approach supports both active recall and contextual understanding.

3.5 Target Audience

The primary target audience of [GAME NAME HERE] is Grade 6 students aligned with the MATATAG curriculum. At this stage, learners are introduced to foundational chemistry concepts such as states of matter, physical and chemical changes, and mixtures. The game's visual style, feedback systems, and mobile-first design make it accessible to a wide range of learners who benefit from interactive and gamified instruction.

3.6 Game Flow Summary

The game progresses through increasingly challenging levels, beginning in the café location. As players advance, new ingredients, action orbs, and mechanics are introduced. Bubble-shooter levels demonstrate various physical transformations such as melting, freezing, cutting, and crushing. Sorting minigame levels also increase in complexity by introducing a greater variety of materials.

3.7 Look and Feel

The game adopts a cartoon-inspired visual style influenced by titles such as *Animal Crossing: New Horizons*, *Kirby's Dream Buffet*, and *Hello Kitty Island Adventure*. Design inspiration is also drawn from PopMart blind box figures, aligning with the game's whimsical and approachable aesthetic. All assets are 2D, stylized, and rounded to emphasize the game's bubbly theme.

3.8 How does the game insert itself in a pedagogical scenario?

[GAME NAME HERE] integrates into pedagogical scenarios by aligning gameplay mechanics with core chemistry learning competencies. It functions as a supplementary instructional tool that reinforces classroom learning through interactive engagement. Teachers may use the game to introduce or review concepts related to physical and chemical changes without the constraints of a physical laboratory. Each level functions as a short lesson paired with a problem-solving activity, providing objectives, challenges, and immediate feedback.

Beyond the bubble-shooter mechanics, the village environment enables contextual application of learned concepts. Players interact with characters, explore locations, and encounter progressively challenging scenarios that connect chemistry concepts to familiar situations. This narrative-driven exploration promotes curiosity and reinforces understanding through storytelling.

The game is designed around two learning objectives aligned with Bloom's Taxonomy.

LO1 (Lower-Order): Classify physical changes and chemical changes by identifying

states of matter and recognizing whether an action results in a physical or chemical transformation, including examples of uniform and non-uniform mixtures.

LO2 (Higher-Order): Explain and justify whether a change is physical or chemical by applying evidence from simulated real-world scenarios, such as predicting outcomes of heating or cooling materials and selecting appropriate actions to achieve desired results.

Through the Village and Bubble Shooter Lab components, [GAME NAME HERE] supports inquiry, experimentation, and critical thinking. Immediate feedback mechanisms, including character reactions and post-level explanations, help correct misconceptions and reinforce accurate understanding.

3.9 Gameplay and Mechanics

This section will be talking about the gameplay and mechanics that are going to be shown by the game [GAME NAME HERE]. This section will talk about how the game is played and how the game will supply the necessary learning objectives to the players.

3.9.1 Gameplay

Core gameplay takes place in a café setting where players complete chemistry-based challenges focused on changes in states of matter, physical and chemical changes, and the effects of heat on materials. Players launch action orbs such as Heat, Cool, Crush, and Cut toward target materials to trigger appropriate reactions. Incorrect actions result in penalties accompanied by visual and auditory feedback. As players progress, challenges increase in complexity through multi-step reactions and environmental modifiers, with guidance provided to support understanding.

3.9.2 Moment to Moment

Phase	Description
-------	-------------

Table 3.1: Game Loop Table

3.9.3 Mechanics

Set within a narrative-driven environment, players complete tasks inspired by everyday situations, such as food preparation and material sorting. Action orbs simulate scientific processes in an accessible arcade-style format. Progress is visualized through a traffic-light system and a three-star scoring mechanic, enabling players to easily assess their performance.

3.9.4 Physics

The game employs basic bubble-shooter physics to simulate predictable trajectories and collisions. These mechanics support intuitive gameplay while reinforcing timing and spatial reasoning skills.

3.9.5 Movement

[GAME NAME HERE] uses a top-down perspective for both gameplay and exploration. A virtual joystick enables movement and aiming, allowing seamless control across mobile and desktop platforms.

3.9.6 Objects

Interactive objects include action orbs, reaction tiles, collectible tokens, compound-forming elements, and village features such as character kiosks and informational displays. Each object supports learning, progression, or narrative development.

3.9.7 Actions

Player actions include aiming and shooting action orbs, switching between available orbs, interacting with characters, unlocking new areas, and viewing educational information within the village.

3.9.8 Combat

The game does not include traditional combat. Instead, challenges are presented as puzzles involving resource management, timing, and condition-based problem solving.

3.9.9 Economy

Gameplay emphasizes resource management. Players must efficiently use limited material orbs to complete level objectives, encouraging critical thinking and strategic planning.

3.9.10 Game Options

[GAME NAME HERE] includes adjustable sound and control sensitivity settings, a contextual help menu, and a pause menu. These options enhance usability and player comfort without interfering with educational objectives.

3.9.11 Constraints due to the pedagogical objectives

To maintain pedagogical integrity, gameplay mechanics are intentionally simplified and scaffolded. All reactions align with Grade 6 learning outcomes, and pacing is designed to minimize stress while reinforcing understanding.

3.10 Story, Setting and Character

3.10.1 Story and Narrative

The narrative follows a chemist assisting in the opening of a neighborhood café. Through completing recipes and tasks, the player observes how ingredients undergo physical and chemical changes. Later challenges introduce irreversible chemical changes, reinforcing key distinctions between transformation types and encouraging reflection on real-world applications.

3.10.2 Game progression

Progression is structured to introduce new concepts gradually, allowing players time to practice and internalize information before advancing. Narrative and gameplay progression are designed to support continuous learning without cognitive overload.

3.10.3 License considerations

All third-party and open-source assets will be properly licensed and credited. In-house code will follow MIT licensing, while creative assets will use Creative Commons Attribution-NonCommercial 4.0 (CC BY-NC 4.0).

3.10.4 Game World

The game world centers on a chemistry-themed café presented in a stylized 2D environment. The setting blends cozy design with experimental elements, providing a playful space for discovery, experimentation, and problem solving.

3.10.5 Characters

Players assume the role of a research assistant guided by the Local Bubble Chemist. Through completing tasks and experiments, players unlock new mechanics and increasingly complex challenges.

3.10.6 Levels

Levels follow a classic bubble-shooter structure inspired by games such as Zuma, combined with exploration and interaction reminiscent of life-simulation games. Friendly guidance and narrative context support learning progression.

3.10.7 Training levels

The training level introduces the game's mechanics and educational goals in a low-pressure environment. Visual cues, simple prompts, and immediate feedback build player confidence and prepare learners for more complex challenges.

3.10.8 Assessment

Player analytics will be integrated to track engagement and learning-related metrics such as playtime and level completion. These data will support analysis of learning progress and engagement.

3.11 Interface

3.11.1 Visual System

A top-down view ensures clear visibility of the play area. Interface elements are minimal and designed to present essential information without obstructing gameplay.

3.11.2 Control System

The game uses screen-based controls optimized for both mouse and touchscreen input, ensuring accessibility across platforms.

3.11.3 Audio, music, sound effects

Audio assets will be sourced from in-house production, open-source libraries, and licensed third-party resources, with proper attribution provided.

3.11.4 Help System

The in-game help system provides tutorials, definitions, and visual examples accessible from the main menu and pause screen, allowing learners to review concepts at their own pace.

CHAPTER 4

DESIGN AND IMPLEMENTATION

4.1 Design Changes

4.2 Implementation Challenges

4.3 Formative User Testing Observations

Formative user testing was conducted with Grade 6 students from Inchican Elementary School using a playable prototype that included the Tutorial Level, Bubble Shooter Level 1, and the Sorting Mini Game. Testing was conducted face-to-face, allowing direct observation of player behavior, engagement, and comprehension. Students demonstrated an overall understanding of gameplay mechanics and responded positively to visual cues and immediate feedback, despite their current science curriculum focusing on topics outside states of matter.

Feedback indicated that some levels were perceived as too easy and quickly memorized, suggesting a need for increased challenge variety. Students also commented on improving visual consistency in the sorting minigame and adding more objects to extend engagement. Language accessibility emerged as a key consideration, with requests for Filipino or Tagalog text. Overall, students expressed enjoyment of the game experience and interest in playing a completed version. These observations will inform future refinements to difficulty balancing, visual design, and language support.

CHAPTER 5

METHODOLOGY

5.1 Research Design

This study employs a mixed-methods research approach. Qualitative data will be collected through one-on-one interviews conducted during and after game development to identify student preferences and evaluate enjoyment. Quantitative data will be gathered through pretests and posttests administered over one month to assess learning gains related to changes in the state of matter.

5.2 Data Collection

Primary data will be collected from interviews and pretest and posttest results involving 20 Grade 6 students in a face-to-face setting. Each researcher will interview five students, with interviews audio-recorded for transcription and analysis. Pretests will assess baseline knowledge through a 25-item quiz administered prior to gameplay. Posttests of equal length will be administered following gameplay sessions monitored weekly by the research team.

Secondary data will include in-game analytics designed to track engagement and progression. These data will be collected ethically, stored securely, and used to supplement findings from tests and interviews.

5.3 Data Analysis

Repeated measures analysis of variance (ANOVA) will be used to compare pretest and posttest scores (DataTab Team, 2025). Qualitative interview data will be analyzed using thematic analysis, involving transcription, coding, theme identification, and interpretation to synthesize findings (Hecker & Kalpokas, 2025).

5.4 Ethical Considerations

Informed consent will be obtained from all participants and their parents or guardians. Participant anonymity will be maintained, and students may withdraw at any time without penalty. Potential risks, such as content misalignment with classroom instruction, will be mitigated through collaboration with teachers and iterative playtesting. All participant data will be securely disposed of upon completion of the study.

REFERENCES

- Barrot, J. S., Llenares, I. I., & del Rosario, L. S. (2021). Students' online learning challenges during the pandemic and how they cope with them: The case of the philippines. *Education and Information Technologies*, 26, 7321–7338. <https://doi.org/10.1007/s10639-021-10589-x>
- Blyznyuk & Kachak. (2024). Title tbd [TODO: Add complete citation details].
- Bridges. (2015). Title tbd [TODO: Add complete citation details].
- Byusa, E., Kampire, E., & Mwesigye, A. R. (2022). Game-based learning approach on students' motivation and understanding of chemistry concepts: A systematic review of literature. *Heliyon*, 8(5). <https://doi.org/10.1016/j.heliyon.2022.e09541>
- DataTab Team. (2025). Repeated measures anova. <https://datatab.net/tutorial/anova-with-repeated-measures>
- Department of Education. (2024). Matatag curriculum guides. <https://www.deped.gov.ph/matatag-curriculum/>
- Diquito, T. (2024). Basic education curriculum under the newly implemented k to 10 (matatag) curriculum in the philippines: The case of science education. *American Journal of Education and Technology (AJET)*, 3(3), 123–132. <https://doi.org/10.54536/ajet.v3i3.3396>
- Hecker, J., & Kalpokas, N. (2025, February). Thematic analysis in mixed methods approach. <https://atlasti.com/guides/thematic-analysis/thematic-analysis-mixed-methods>
- Hu, Y., Gallagher, T., Wouters, P., van der Schaaf, M., & Kester, L. (2022). Game-based learning has good chemistry with chemistry education: A three-level meta-analysis. *Journal of Research in Science Teaching*, 59(9), 1499–1543. <https://doi.org/10.1002/tea.21765>
- Janah, M., & Mitarlis, M. (2024). Development of chemistry learning media based on joyful learning with snakes and ladders game on stoichiometry material to improve student learning outcomes. *Hydrogen: Jurnal Kependidikan Kimia*, 12(1), 135–150. <https://doi.org/10.33394/hjkk.v12i1.10192>
- Kilag, O. K., Andrin, G., Abellanos, C., Villaver, M., Uy, F., & Sasan, J. M. (2024). K-12 program challenges. *RiseJournals.org*, 1(5), 172–177. <https://doi.org/10.5281/zenodo.11183037>
- Maung, K. (2015). Bias blaster: Aiding cognitive bias modification-interpretation through a bubble shooter induced game-flow. https://web.archive.org/web/20190221022335id_/http://pdfs.semanticscholar.org/2b97/8230e43e7c75cbb19ebc9cfd38fc0921f47e.pdf
- Putri, I. Y. V. S., Rahayu, S., & Dasna, I. W. (2022). Game-based learning application in chemistry learning: A systematic literature review. *Jurnal Pendidikan MIPA*, 23(1), 1–12. <https://doi.org/10.23960/jpmipa/v23i1.pp01-12>
- Ren, J., Xu, W., & Liu, Z. (2024). The impact of educational games on learning outcomes: Evidence from a meta-analysis. *International Journal of Game-Based Learning (IJGBL)*, 14(1), 1–25. <https://doi.org/10.4018/IJGBL.336478>

- Saputri, Pranata, & Zebua. (2025). Title tbd [TODO: Add complete citation details].
- Sison, R. C., Espulgar, C., Madrigal, L., Obedoza, J., Romblon, N., Sace, J., & Si, M. (2018). Toward an outcome-based methodology for developing game-based learning environments. *Proceedings of the 26th International Conference on Computers in Education*, 613–618. https://animorepository.dlsu.edu.ph/faculty_research/1054
- Treagust, Duit, & Nieswandt. (2018). Title tbd [TODO: Add complete citation details].
- Vallespin. (2024). Title tbd [TODO: Add complete citation details].
- Vega. (2024). Title tbd [TODO: Add complete citation details].